Gujarat University

Department of Computer Science

MCA Semester 3

Computer Vision (MCA216(1A))

Theory Assignment: Unit 5 - Motion Estimation

Last Date of Submission: 15/10/2026

Answer all the questions.

1. What is Motion Analysis in Computer Vision? Explain its importance and applications.
2. What is Activity Recognition? Explain the different types of human activities that can be recognized using Computer Vision.
3. Explain the difference between Motion Detection, Motion Tracking, and Activity Recognition.
4. What is Motion Detection? Explain different techniques used for detecting motion in video sequences.
5. Explain the Background Subtraction technique for motion detection.
6. What is Optical Flow? Explain its concept and applications in motion analysis.
7. Explain Motion Tracking in Computer Vision. What are the major challenges involved in tracking objects?
8. Explain different object tracking techniques used in Computer Vision.
9. What is the difference between single-object tracking and multiple-object tracking?
10. Explain the concept of human activity recognition from video/image sequences.
11. How can human activities be inferred from a sequence of images? Explain the major steps involved.
12. Explain the role of features such as shape, motion, and appearance in human activity recognition.
13. Explain the use of machine learning and deep learning for human activity recognition.
14. Discuss the applications of motion detection and human activity recognition in surveillance, healthcare, and sports.
15. Explain the challenges and limitations of inferring human activities from image sequences.